

# FOUNDATION\_PACKAGES

## Foundation Packages (pkg/)

This catalogue describes the pure-Go foundation library that the Helix control plane is built from. There are **126+ packages** under pkg/ (plus 19 private subsystems under internal/).

Every foundation package follows the same engineering contract:

- **Pure Go, standard-library-first.** External dependencies are avoided; where an in-repo primitive is reused it is imported read-only.
- **Deterministic.** Clocks, randomness, and I/O sources are injected (interfaces / seeds / explicit ticks) so tests are reproducible. No `time.Now()` inside logic under test.
- **Mutation-proven (CLAUDE-1).** Each package's tests are designed to *fail* if the specific logic they cover is mutated — green tests over stubs or always-true conditions are forbidden. Items are not marked complete without an independent mutation bite confirming the guard is live.
- **Cross-platform parity (CLAUDE-2).** OS-specific capabilities use real native facilities per platform; there are no Linux-only mocks behind build tags for real-operation features.

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## Consensus & Coordination

| Package                     | Purpose                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| voting                      | Largest-subcluster-wins quorum (3/2 survive+fence, 2/2 both-fence, TTL re-resolve).                                               |
| failconfirm                 | Redis-cluster two-phase failure confirmation (PFAIL→FAIL via distinct-reporter majority quorum, OnDead-once).                     |
| heartbeatcoalescer          | Multi-Raft per-peer heartbeat coalescing, $O(\text{shards} \cdot \text{peers}) \rightarrow O(\text{peers})$ , liveness preserved. |
| leader, lock                | Leadership and advisory locking primitives.                                                                                       |
| splitbrain, splitbrainalert | Split-brain detection and alerting.                                                                                               |

| Package      | Purpose                                                                                                                                                         |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| phasesgate   | Sub-phase dependency-gate validation.                                                                                                                           |
| epochresolve | Config-epoch failover slot-ownership conflict resolution (highest-epoch wins, deterministic lexicographic equal-epoch tiebreak, order-independent convergence). |

## Membership & Discovery

| Package                           | Purpose                                                                                                                                  |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| swim                              | SWIM gossip membership with phi-accrual failure detection and hierarchical tiers.                                                        |
| discovery (+ discovery/federated) | Service discovery, including federated cross-cell discovery.                                                                             |
| nattraversal                      | STUN-based NAT traversal.                                                                                                                |
| ice                               | ICE connectivity establishment.                                                                                                          |
| cellmesh                          | Cross-cell mesh networking.                                                                                                              |
| scan                              | Oracle-SCAN-style stable virtual client endpoint: least-loaded healthy-backend routing with a membership-stable name and injected clock. |

## Replication & State

| Package                       | Purpose                                                                                                  |
|-------------------------------|----------------------------------------------------------------------------------------------------------|
| crdt (+ crdt/merkle)          | LWW / OR-Set / G-Counter / PN-Counter / vector-clock CRDTs and Merkle anti-entropy trees.                |
| mvcc                          | MVCC revision store with a real B-tree index, time-travel reads, watch-from-revision, and compaction.    |
| antientropy                   | Cassandra-style 3-layer repair: hinted handoff, read repair, Merkle-tree diff (over in-memory replicas). |
| watchmanager                  | etcd-style persistent watch manager with synced / unsynced / victim watcher groups.                      |
| hlc                           | Hybrid logical clocks.                                                                                   |
| offlinesync, checkpoint_merge | Offline delta sync and checkpoint merging.                                                               |
| tieredcache                   | Hot(memory)/warm(NVMe)/cold(SSD) tiered cache with injected backends + injected clock, idle-             |

| Package   | Purpose                                                                                                                                                                                                                        |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| deltacrdt | demotion + promotion, and per-tier hit stats.<br><br>Standalone delta-state CRDTs (G-counter / PN-counter / observed-remove OR-set / LWW-map) with delta-mutators and a convergent (commutative/associative/idempotent) merge. |

## Scheduling & Placement

| Package          | Purpose                                                                                                                      |
|------------------|------------------------------------------------------------------------------------------------------------------------------|
| scheduler        | Omega-model two-level scheduler: ClassAd matching, gang scheduling (all-or-nothing), preemption, optimistic concurrency.     |
| constraints      | Pacemaker-style four-type constraint engine (location / colocation / order / stickiness).                                    |
| preempt          | Value-multiplier (Gepetto) preemption with sole-global protection.                                                           |
| priorityqueue    | Multifactor composite priority (age + fairshare + size + QoS) with clock-driven aging.                                       |
| backfill         | SLURM-style conservative backfill over an availability timeline.                                                             |
| workclaim        | SKIP-LOCKED-style optimistic work-claiming (exactly-once under contention).                                                  |
| admissioncontrol | N+K capacity-reserve failover admission control.                                                                             |
| budgetcap        | Global MaxCostPerHour budget cap on allocations.                                                                             |
| qos              | QoS-tier routing (real-time / interactive / batch / best-effort).                                                            |
| suitability      | HPC-vs-inference workload classifier and placement routing.                                                                  |
| ewmarank         | Utilization-aware EWMA candidate ranking with sticky-client routing.                                                         |
| fallbackchain    | Per-task ordered fallback chains with dedup, attempt cap, and empty-result handling.                                         |
| providerchain    | Ordered multi-tier provider fallback cascade with retrievable/terminal error classification and injected-clock failover SLA. |

| Package     | Purpose                                                                                                                                                         |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| modelrouter | Strategy-to-default-model resolution (latency / throughput / quality / cost) with a typed unknown-strategy error.                                               |
| gepetto     | HelixGepetto dual-resource local-vs-Chutes arbitration with a monotonic reserve schedule and a strict $>0.80$ high-water anti-starvation zero-capacity cut-off. |
| smartrouter | Intelligent ModelRouter (latency / throughput / cost / TEE / balanced) that always excludes unhealthy models, with a typed no-eligible-model error.             |
| llmfailover | Error-classified LLM failover taxonomy (deterministic per-class fallback paths) plus a sandbox scheduling-hint contract.                                        |

## GPU, Resource & Cost Management

| Package                                      | Purpose                                                                                                                                                                       |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pool                                         | Node/instance pool manager with a GPUProvider provisioning seam.                                                                                                              |
| local                                        | Local GPU TCO model (amortized capital + power, utilization-scaled).                                                                                                          |
| costsched, latencysched                      | Cost-aware and latency-aware schedulers.                                                                                                                                      |
| healthmonitor                                | Edge-triggered health transitions with auto-failover / recovery callbacks.                                                                                                    |
| healthprobe                                  | Startup / readiness / liveness probe tiers with a startup grace period.                                                                                                       |
| gpuattest                                    | Device attestation crypto: challenge/response + fingerprint, seeded-matmul proof-of-GPU-work, $O(1)$ Merkle spot-check, device-sealed AES-GCM/HKDF.                           |
| attestadmit                                  | Attestation-gated scheduler admission predicate (uses gpuattest).                                                                                                             |
| quantization                                 | Per-tier model-variant selection (memory-fit + backend-support gates).                                                                                                        |
| capability, deviceprofile, device, devicemap | Capability negotiation and device profiling.                                                                                                                                  |
| devicecatalog                                | Machine-readable 64-device taxonomy with case-insensitive substring Lookup resolving a discovered CPU model (e.g. Rockchip RK3588) to its tier / trust-level / compute-class. |
| tierdef, tiersec                             | Tier definitions and tier security.                                                                                                                                           |

| Package                       | Purpose                                                                                                                                                                                                                   |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| burst, cloudspot, marketplace | Burst-to-cloud autoscaling, spot pricing, and compute marketplace.                                                                                                                                                        |
| revenueopt                    | RevenueOptimizer.OptimizeAllocation greedy GPU→marketplace assignment maximising expected revenue, with a TEE→Chutes revenue bias.                                                                                        |
| carbonsched                   | Carbon-aware job placement: selects the lowest-carbon-intensity latency-eligible region (deterministic name tiebreak) with per-job kWh / gCO2 metering.                                                                   |
| workloadrouter                | UnifiedManager.RouteWorkload: concurrent per-marketplace price probes + a weighted composite score (inverse price/latency, health) with a TEE multiplier re-ranking confidential offers when the workload requires a TEE. |
| economics                     | RewardDistributor multi-token distribution with treasury/reinvest splits and a conservation invariant, plus GetParticipantROI (ROI% + break-even days, typed unknown-participant error).                                  |
| bursthysteresis               | BurstController with a MONITOR→SPILL→RECOVER two-threshold hysteresis dead-band.                                                                                                                                          |
| gpucatalog                    | SUPPORTED_GPUS compute-multiplier catalog, LookupMultiplier, and attestation-aware Score (attested scores rank above un-attested).                                                                                        |
| gputopo                       | Topology-aware NUMA/NVLink GPU placement scoring that prefers NVLink-connected GPU pairs and falls back to PCIe.                                                                                                          |
| balancemonitor                | USD account-balance monitor with a strict low-balance floor warning, over HTTP with Authorization: Bearer.                                                                                                                |

## Federation & Multi-cluster

| Package                             | Purpose                                  |
|-------------------------------------|------------------------------------------|
| federation (+ federation/suspicion) | Federation membership and suspicion.     |
| fedtopology                         | Federation topology patterns.            |
| fedtrust                            | Cross-cluster federated-trust admission. |
| configsync                          | CRDT (LWW) config synchronisation.       |
| residency                           | Data-residency admission evaluation.     |
| raftprofile                         | etcd Raft WAN tuning profiles.           |

| Package                | Purpose                                                |
|------------------------|--------------------------------------------------------|
| spiffefed, doublecrypt | SPIFFE federation (mTLS + x509) and double-encryption. |

## Messaging, Flow & Routing

| Package                      | Purpose                                                                        |
|------------------------------|--------------------------------------------------------------------------------|
| flowcontrol                  | Kubernetes API Priority & Fairness (FlowSchema → PriorityLevel → fair queues). |
| workqueue                    | Rate-limited work queue with exponential backoff and dedup.                    |
| ratelimit, backoff, retry    | Rate limiting, exponential backoff, retry.                                     |
| idempotent                   | Exactly-once producer via producer-id + sequence.                              |
| rebalance                    | Kafka cooperative-sticky incremental partition rebalancing.                    |
| informer                     | Informer-style list-watch local cache with indexers and resync.                |
| redundantexec                | BOINC-style redundant-execution trust scorer.                                  |
| hashslot                     | CRC16 16,384 hash-slot router with MOVED/ASK in-flight migration redirection.  |
| slotmigration                | Atomic live-session slot migration FSM (PREPARE → TRANSFER → COMMIT/ABORT).    |
| fiber, pubsub, events, serde | Framed transport, pub/sub, events, serialization.                              |

## Edge

| Package      | Purpose                                                                           |
|--------------|-----------------------------------------------------------------------------------|
| edge         | Edge restriction, schedule rules, trust, and work units.                          |
| edgeregistry | Multi-tier (T3–T8) edge device registration with tier/trust/capability labels.    |
| edgeverify   | Redundant edge-output verification against a trusted anchor via SHA-256 checksum. |
| edgefusion   | Edge data fusion.                                                                 |

## Security & Verification

| Package                               | Purpose                                                                                                                                                                                                                                                                 |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| crypto, jwt                           | Cryptographic primitives and JWT auth.                                                                                                                                                                                                                                  |
| modelintegrity                        | HF-cache verification gate (SHA-256 + size).                                                                                                                                                                                                                            |
| gravaladmit                           | HMAC-SHA256 GraVal challenge-response provider admission gate with single-use nonce and a <code>graval.verified</code> label.                                                                                                                                           |
| gravalverify                          | GraVal GPU attestation with a VRAM-ratio $\geq 0.95$ gate and a concurrent, race-clean BatchVerify pass-rate KPI.                                                                                                                                                       |
| fsresidency                           | Filesystem residency challenge: per-byte-range ReadAt + SHA-256 verification proving a model file is really present on disk; truncated / missing / mismatch all fail.                                                                                                   |
| exportcontrol                         | Export-control / country-code KYC gate on node onboarding: a controlled GPU (H100/A100) from a Tier-3 (embargoed) or unknown jurisdiction is rejected with <code>ErrExportDenied</code> ; Tier-1 is allowed.                                                            |
| hybridkex                             | Hybrid post-quantum key exchange combining X25519 (crypto/ecdh) + ML-KEM-768 (crypto/mlkem) shared secrets via HKDF-SHA256 (negotiated group X25519MLKEM768); both parties derive an identical 32-byte key, and a tampered ML-KEM ciphertext yields a non-matching key. |
| compliancecdoc                        | EU AI Act compliance-documentation pipeline: generates a model card + technical-documentation artifact from attestation-log records with a SHA-256 provenance hash bound to the record content; empty logs $\rightarrow$ <code>ErrNoAttestations</code> .               |
| gpuattest (multigpu.go)               | Multi-GPU node enumeration: an N-GPU node descriptor yields N distinct, independently-verifiable per-device attestation proofs keyed by GPU UUID; rejects duplicate/empty descriptors.                                                                                  |
| redundantexec, attestadmit, gpuattest | See above — redundancy trust, attestation admission, attestation crypto.                                                                                                                                                                                                |

## Testing, Simulation & Quality

| Package                                 | Purpose                                                                                                |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------|
| testing/dst (+ BUGGIFY, chaos, turmoil) | FoundationDB-style deterministic simulation testing, BUGGIFY fault macros, network simulation.         |
| timefault                               | Clock-fault injectors (skew / freeze / monotonic-violation) + skew detector for split-brain avoidance. |
| chaosexp                                | Chaos experiment suite with delivery-level steady-state.                                               |
| fmea                                    | FMEA catalogue + RPN validator.                                                                        |
| phasegate, phase7matrix                 | Phase dependency gates and the Phase-7 gap-matrix verifier.                                            |
| qualitygate, covgate, stats, sandbox    | Quality gates, coverage gates, statistics (Welch t-test), sandboxing.                                  |

## External Provider Integration

| Package       | Purpose                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------|
| chutesaccount | Chutes API client over HTTP (Authorization: Bearer): model-list (TEE / price fields) and /users/me account-balance retrieval.      |
| llmadapter    | Claude Messages / OpenAI Responses-API request and response shape adapters to OpenAI chat-completions, preserving tool-call order. |

## Observability

| Package     | Purpose                                                                                                                                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| metrics     | Metrics collection (Prometheus-text NodeCollector); plus TierMetrics (tiermetrics.go) exposing gpu_tier_utilization, gpu_cost_per_hour, and provider_health series with deterministic ordering and concurrent-safe setters. |
| tracing     | W3C distributed tracing.                                                                                                                                                                                                    |
| health      | Health aggregation.                                                                                                                                                                                                         |
| grafanadash | Grafana dashboard generation + validation.                                                                                                                                                                                  |



| Package | Purpose             |
|---------|---------------------|
| log     | Structured logging. |

*This catalogue is maintained as the foundation library grows. For the authoritative list run `ls pkg/`. For the work-item registry that tracks package delivery, see [HXC\\_REGISTRY.md](#).*